MPG No. 7.1.20 $mm2$ [Type I, orthorhombic] [Q tensor]

* Rank 0 tensor.

$$[Q]$$

$$Q = \mathbb{Q}_0^{(1)}$$

* Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & Q_z \end{bmatrix}$$

$$Q_z = \mathbb{Q}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} Q_{xx} & 0 & 0 \\ 0 & Q_{yy} & 0 \\ 0 & 0 & Q_{zz} \end{bmatrix}$$

$$Q_{xx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(1)} + \mathbb{Q}_v^{(1)}$$
$$Q_{yy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(1)} - \mathbb{Q}_v^{(1)}$$
$$Q_{zz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & Q_{xxz} \\ 0 & 0 & Q_{yyz} \\ 0 & 0 & Q_{zzz} \\ 0 & Q_{yyz} & 0 \\ Q_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxz} = \mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{\alpha(1)} + \mathbb{Q}_z^{\beta(1)}$$
$$Q_{yyz} = \mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{\alpha(1)} - \mathbb{Q}_z^{\beta(1)}$$
$$Q_{zzz} = 3\mathbb{Q}_z^{(1)} + 2\mathbb{Q}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & Q_{xxz} \\ 0 & 0 & Q_{yyz} \\ 0 & 0 & 0 \\ 0 & -\frac{Q_{yyz}}{2} & 0 \\ -\frac{Q_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxz} = 2\mathbb{G}_{xy}^{(1)} + 2\mathbb{Q}_z^{(2)}$$
$$Q_{yyz} = -2\mathbb{G}_{xy}^{(1)} + 2\mathbb{Q}_z^{(2)}$$
$$\frac{Q_{yyz}}{2} = -\mathbb{G}_{xy}^{(1)} + \mathbb{Q}_z^{(2)}$$
$$\frac{Q_{xxz}}{2} = \mathbb{G}_{xy}^{(1)} + \mathbb{Q}_z^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & Q_{yzy} & 0 \\ Q_{zxx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{yzy} = \mathbb{G}^{(2)}_{xy} + \mathbb{Q}^{(3)}_{z}$$
$$Q_{zxx} = \mathbb{G}^{(2)}_{xy} - \mathbb{Q}^{(3)}_{z}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & 0 & 0 \\ Q_{xxyy} & Q_{yyyy} & Q_{yyzz} & 0 & 0 & 0 \\ Q_{xxzz} & Q_{yyzz} & Q_{zzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{zxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}^{(1)}_{0} + 2\mathbb{Q}^{(2)}_{0} - \mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4v} + 2\mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(1)}_{u} - 4\mathbb{Q}^{(2)}_{u} + 2\mathbb{Q}^{(1)}_{v} + 4\mathbb{Q}^{(2)}_{v}$$
$$Q_{xxyy} = \mathbb{Q}^{(1)}_{0} + 2\mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(1)}_{u}$$
$$Q_{xxzz} = \mathbb{Q}^{(1)}_{0} - \mathbb{Q}^{(1)}_{4u} + \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(1)}_{u} + \mathbb{Q}^{(1)}_{v}$$
$$Q_{yyyy} = \mathbb{Q}^{(1)}_{0} + 2\mathbb{Q}^{(2)}_{0} - \mathbb{Q}^{(1)}_{4u} + \mathbb{Q}^{(1)}_{4v} + 2\mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(1)}_{u} - 4\mathbb{Q}^{(2)}_{u} - 2\mathbb{Q}^{(1)}_{v} - 4\mathbb{Q}^{(2)}_{v}$$
$$Q_{yyzz} = \mathbb{Q}^{(1)}_{0} - \mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(1)}_{u} - \mathbb{Q}^{(1)}_{v}$$
$$Q_{zzzz} = \mathbb{Q}^{(1)}_{0} + 2\mathbb{Q}^{(2)}_{0} + 2\mathbb{Q}^{(1)}_{4u} + 2\mathbb{Q}^{(1)}_{4} + 4\mathbb{Q}^{(1)}_{u} + 8\mathbb{Q}^{(2)}_{u}$$
$$Q_{yzyz} = \mathbb{Q}^{(2)}_{0} - \mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(2)}_{u} - \mathbb{Q}^{(2)}_{v}$$
$$Q_{zxzx} = \mathbb{Q}^{(2)}_{0} - \mathbb{Q}^{(1)}_{4u} + \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(2)}_{u} + \mathbb{Q}^{(2)}_{v}$$
$$Q_{xyxy} = \mathbb{Q}^{(2)}_{0} + 2\mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(2)}_{u}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & Q_{xxyy} & Q_{xxzz} & 0 & 0 & 0 \\ -Q_{xxyy} & 0 & Q_{yyzz} & 0 & 0 & 0 \\ -Q_{xxzz} & -Q_{yyzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxyy} = 4\mathbb{G}^{(1)}_{xyz} - 2\mathbb{Q}^{(3)}_{v}$$
$$Q_{xxzz} = -4\mathbb{G}^{(1)}_{xyz} + 3\mathbb{Q}^{(3)}_{u} - \mathbb{Q}^{(3)}_{v}$$
$$Q_{yyzz} = 4\mathbb{G}^{(1)}_{xyz} + 3\mathbb{Q}^{(3)}_{u} + \mathbb{Q}^{(3)}_{v}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} Q_{yzyz} & 0 & 0 \\ 0 & Q_{zxzx} & 0 \\ 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{Q}^{(3)}_{0} - 2\mathbb{Q}^{(4)}_{u} + 2\mathbb{Q}^{(4)}_{v}$$
$$Q_{zxzx} = \mathbb{Q}^{(3)}_{0} - 2\mathbb{Q}^{(4)}_{u} - 2\mathbb{Q}^{(4)}_{v}$$
$$Q_{xyxy} = \mathbb{Q}^{(3)}_{0} + 4\mathbb{Q}^{(4)}_{u}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ Q_{yzyz} & 0 & 0 \\ 0 & Q_{zxzx} & 0 \\ 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{G}^{(2)}_{xyz} - 3\mathbb{Q}^{(5)}_{u} - \mathbb{Q}^{(5)}_{v}$$
$$Q_{zxzx} = \mathbb{G}^{(2)}_{xyz} + 3\mathbb{Q}^{(5)}_{u} - \mathbb{Q}^{(5)}_{v}$$
$$Q_{xyxy} = \mathbb{G}^{(2)}_{xyz} + 2\mathbb{Q}^{(5)}_{v}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{zxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{G}^{(3)}_{xyz} - 3\mathbb{Q}^{(6)}_{u} - \mathbb{Q}^{(6)}_{v}$$
$$Q_{zxzx} = \mathbb{G}^{(3)}_{xyz} + 3\mathbb{Q}^{(6)}_{u} - \mathbb{Q}^{(6)}_{v}$$
$$Q_{xyxy} = \mathbb{G}^{(3)}_{xyz} + 2\mathbb{Q}^{(6)}_{v}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ Q_{yyxx} & Q_{yyyy} & Q_{yyzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ Q_{zzxx} & Q_{zzyy} & Q_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & Q_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{zxzx} & 0 & 0 & Q_{zxxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{xyxy} & 0 & 0 & Q_{xyyx} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}^{(1)}_{0} + 2\mathbb{Q}^{(2)}_{0} - \mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4v} + 2\mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(1)}_{u} - 4\mathbb{Q}^{(2)}_{u} + 2\mathbb{Q}^{(1)}_{v} + 4\mathbb{Q}^{(2)}_{v}$$
$$Q_{xxyy} = 4\mathbb{G}^{(1)}_{xyz} + \mathbb{Q}^{(1)}_{0} + 2\mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(1)}_{u} - 2\mathbb{Q}^{(3)}_{v}$$
$$Q_{xxzz} = -4\mathbb{G}^{(1)}_{xyz} + \mathbb{Q}^{(1)}_{0} - \mathbb{Q}^{(1)}_{4u} + \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(1)}_{u} + 3\mathbb{Q}^{(3)}_{u} + \mathbb{Q}^{(1)}_{v} - \mathbb{Q}^{(3)}_{v}$$
$$Q_{yyxx} = -4\mathbb{G}^{(1)}_{xyz} + \mathbb{Q}^{(1)}_{0} + 2\mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(1)}_{u} + 2\mathbb{Q}^{(3)}_{v}$$
$$Q_{yyyy} = \mathbb{Q}^{(1)}_{0} + 2\mathbb{Q}^{(2)}_{0} - \mathbb{Q}^{(1)}_{4u} + \mathbb{Q}^{(1)}_{4v} + 2\mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(1)}_{u} - 4\mathbb{Q}^{(2)}_{u} - 2\mathbb{Q}^{(1)}_{v} - 4\mathbb{Q}^{(2)}_{v}$$
$$Q_{yyzz} = 4\mathbb{G}^{(1)}_{xyz} + \mathbb{Q}^{(1)}_{0} - \mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(1)}_{u} + 3\mathbb{Q}^{(3)}_{u} - \mathbb{Q}^{(1)}_{v} + \mathbb{Q}^{(3)}_{v}$$
$$Q_{zzxx} = 4\mathbb{G}^{(1)}_{xyz} + \mathbb{Q}^{(1)}_{0} - \mathbb{Q}^{(1)}_{4u} + \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(1)}_{u} - 3\mathbb{Q}^{(3)}_{u} + \mathbb{Q}^{(1)}_{v} + \mathbb{Q}^{(3)}_{v}$$
$$Q_{zzyy} = -4\mathbb{G}^{(1)}_{xyz} + \mathbb{Q}^{(1)}_{0} - \mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(1)}_{u} - 3\mathbb{Q}^{(3)}_{u} - \mathbb{Q}^{(1)}_{v} - \mathbb{Q}^{(3)}_{v}$$
$$Q_{zzzz} = \mathbb{Q}^{(1)}_{0} + 2\mathbb{Q}^{(2)}_{0} + 2\mathbb{Q}^{(1)}_{4u} + 2\mathbb{Q}^{(1)}_{4} + 4\mathbb{Q}^{(1)}_{u} + 8\mathbb{Q}^{(2)}_{u}$$
$$Q_{yzyz} = \mathbb{G}^{(2)}_{xyz} + \mathbb{Q}^{(2)}_{0} - \mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(2)}_{u} - 3\mathbb{Q}^{(5)}_{u} - \mathbb{Q}^{(2)}_{v} - \mathbb{Q}^{(5)}_{v}$$
$$Q_{yzzy} = -\mathbb{G}^{(2)}_{xyz} + \mathbb{Q}^{(2)}_{0} - \mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(2)}_{u} + 3\mathbb{Q}^{(5)}_{u} - \mathbb{Q}^{(2)}_{v} + \mathbb{Q}^{(5)}_{v}$$
$$Q_{zxzx} = \mathbb{G}^{(2)}_{xyz} + \mathbb{Q}^{(2)}_{0} - \mathbb{Q}^{(1)}_{4u} + \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(2)}_{u} + 3\mathbb{Q}^{(5)}_{u} + \mathbb{Q}^{(2)}_{v} - \mathbb{Q}^{(5)}_{v}$$
$$Q_{zxxz} = -\mathbb{G}^{(2)}_{xyz} + \mathbb{Q}^{(2)}_{0} - \mathbb{Q}^{(1)}_{4u} + \mathbb{Q}^{(1)}_{4v} - \mathbb{Q}^{(1)}_{4} + \mathbb{Q}^{(2)}_{u} - 3\mathbb{Q}^{(5)}_{u} + \mathbb{Q}^{(2)}_{v} + \mathbb{Q}^{(5)}_{v}$$
$$Q_{xyxy} = \mathbb{G}^{(2)}_{xyz} + \mathbb{Q}^{(2)}_{0} + 2\mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(2)}_{u} + 2\mathbb{Q}^{(5)}_{v}$$
$$Q_{xyyx} = -\mathbb{G}^{(2)}_{xyz} + \mathbb{Q}^{(2)}_{0} + 2\mathbb{Q}^{(1)}_{4u} - \mathbb{Q}^{(1)}_{4} - 2\mathbb{Q}^{(2)}_{u} - 2\mathbb{Q}^{(5)}_{v}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & 0 & 0 & Q_{yzzy} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{zxzx} & 0 & 0 & Q_{zxxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{xyxy} & 0 & 0 & Q_{xyyx} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{G}^{(3)}_{xyz} + \mathbb{Q}^{(3)}_{0} - 2\mathbb{Q}^{(4)}_{u} - 3\mathbb{Q}^{(6)}_{u} + 2\mathbb{Q}^{(4)}_{v} - \mathbb{Q}^{(6)}_{v}$$
$$Q_{yzzy} = \mathbb{G}^{(3)}_{xyz} - \mathbb{Q}^{(3)}_{0} + 2\mathbb{Q}^{(4)}_{u} - 3\mathbb{Q}^{(6)}_{u} - 2\mathbb{Q}^{(4)}_{v} - \mathbb{Q}^{(6)}_{v}$$
$$Q_{zxzx} = \mathbb{G}^{(3)}_{xyz} + \mathbb{Q}^{(3)}_{0} - 2\mathbb{Q}^{(4)}_{u} + 3\mathbb{Q}^{(6)}_{u} - 2\mathbb{Q}^{(4)}_{v} - \mathbb{Q}^{(6)}_{v}$$
$$Q_{zxxz} = \mathbb{G}^{(3)}_{xyz} - \mathbb{Q}^{(3)}_{0} + 2\mathbb{Q}^{(4)}_{u} + 3\mathbb{Q}^{(6)}_{u} + 2\mathbb{Q}^{(4)}_{v} - \mathbb{Q}^{(6)}_{v}$$
$$Q_{xyxy} = \mathbb{G}^{(3)}_{xyz} + \mathbb{Q}^{(3)}_{0} + 4\mathbb{Q}^{(4)}_{u} + 2\mathbb{Q}^{(6)}_{v}$$
$$Q_{xyyx} = \mathbb{G}^{(3)}_{xyz} - \mathbb{Q}^{(3)}_{0} - 4\mathbb{Q}^{(4)}_{u} + 2\mathbb{Q}^{(6)}_{v}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} Q_{xxxx} & 0 & 0 \\ 0 & Q_{yyyy} & 0 \\ 0 & 0 & Q_{zzzz} \\ 0 & 0 & Q_{yyzz} \\ Q_{zzxx} & 0 & 0 \\ 0 & Q_{xxyy} & 0 \\ 0 & Q_{yzzy} & 0 \\ 0 & 0 & Q_{zxxz} \\ Q_{xxyy} + Q_{yyzz} - Q_{yzzy} - Q_{zxxz} + Q_{zzxx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxxx} = 3\mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} + 2\mathbb{Q}_v^{(1)} + 4\mathbb{Q}_v^{(2)}$$
$$Q_{yyyy} = 3\mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} - 2\mathbb{Q}_v^{(1)} - 4\mathbb{Q}_v^{(2)}$$
$$Q_{zzzz} = 3\mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$
$$Q_{yyzz} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)}$$
$$Q_{zzxx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)}$$
$$Q_{xxyy} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 2\mathbb{Q}_v^{(3)}$$
$$Q_{yzzy} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)}$$
$$Q_{zxxz} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)} + \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)}$$